Rossini Martyr

Data Science | Analytics | Engineering

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MOTIVATION

I am a professional engineer, driven to **solve business problems** using Data Science & Analytics. I bring a systematic and collaborative approach to **add tangible value** to the team, business, and clients. I am a **life-long learner**, constantly seeking new ways to improve and innovate.

SKILLS & TOOLS

Programming: SQL, Python (Pandas, Numpy, Matplotlib, Scikit-Learn, Langchain)
Tools: Excel, Tableau, Github, Streamlit, AWS SageMaker, AutoML, Folium
Math: Statistics (Hypothesis Testing, AB Testing, Causal Inference), Linear Algebra
Machine Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, k-means, PCA, Gradient Boosted Trees, Retrieval Augmented Generation

PROJECTS

Predicting Property Damage due to Flooding Using ML (Github)

- In order to quantify financial exposure due to flooding, I built and deployed a live, interactive, web application, using Streamlit, that predicts flood damage as a function of property cost. I tuned a **Random Forest regression model** using publicly available insurance claims data to predict the damage ratio to **+/- 15%** supporting risk based insurance premium optimization. I optimized the model parameters using cross validation and used SHAP to determine and explain feature importance.

Quantifying Sales Uplift using Causal Impact Analysis (Github)

- To quantify the financial uplift of a new membership tier, I conducted a **Causal Impact Analysis** to measure if waiving per-order delivery fees increased overall customer spending for a grocery retailer. I compared members against a control group of non-members using three months of pre- and post-campaign transaction data. The analysis revealed a statistically significant (95%) sales uplift of 41.1% among club members, suggesting that eliminating delivery friction drove higher overall revenue.

Building an AI Help-Desk Assistant Using RAG (Github)

- To automate a retailer's busy help-desk, I built a **Retrieval-Augmented Generation (RAG)** assistant to answer operational queries. I engineered the pipeline by chunking internal documentation into dense vector embeddings stored in a persistent database, incorporating conversational memory for multi-turn interactions. **Monitored and evaluated via LangSmith**, the system achieved high precision by strictly grounding answers in retrieved context and deploying fallback responses to eliminate hallucinations.

Assessing Campaign Performance Using A|B Testing (Github)

- To optimize campaign ROI for a new membership tier, I conducted an A/B test to determine if expensive promotional mailers drove higher signup rates than cheaper alternatives. Utilizing a Chi-Square Test for Independence, I compared their conversion rates. The analysis revealed that while the expensive mailer yielded a higher raw signup rate, the difference was not statistically significant. This showed that the expensive marketing materials did not conclusively guarantee higher conversion rates.

WORK EXPERIENCE

Senior Engineer, Fast + Epp (Toronto, Canada) February 2024 – Present

- Project lead for the technical delivery of the $120M New Brunswick Museum project, optimizing structural solutions to save over $200,000 in material and resource efficiency
- Managed cross-functional teams on $66M infrastructure projects, meeting stakeholder requirements while ensuring on-time and on-budget delivery of key deliverables
- Mentored junior and intermediate engineers, providing career guidance and subject matter review

Senior Engineer, Arup Canada (Toronto, Canada) April 2016 – February 2024

- Developed an SQL and Python-based analytical workflow to process and analyze datasets with millions of rows, leveraging Amazon Redshift database and cloud compute power to accelerate and iterate design solutions
- Automated the creation of analytical models of underground stations in Toronto using Python-based API's, improving efficiency and accuracy by approximately 30%
- Supervised multiple junior engineers, conducting annual reviews, reviewing peer feedback and providing recommendations on career direction and promotion readiness

EDUCATION

MASc (Structural/Earthquake Engineering) (GPA: 3.94/4.00) 2013 - 2016
University of Toronto

- *P*ublished M.A.Sc research in prestigious structural engineering journal

BASc (Civil Engineering) (GPA: 3.88/4.00) 2008 - 2013
University of Toronto

- Graduated #1 in Civil Engineering

COURSES, CERTIFICATES AND QUALIFICATIONS

Data Science & AI Professional Certification 2026 - Ongoing
Data Science Infinity

- **Actionable Learnings:** Advanced data visualizations using Tableau. Designing, executing, and evaluating rigorous A/B tests and causal impact analyses to measure business impact and drive data-informed product decisions. Packaging machine learning models into production-ready APIs and deploying them to AWS cloud environments using SageMaker and AutoML.

Data Science Professional Certification 2025 - 2026
IBM

- **Actionable Learnings:** Querying databases and manipulating data structures using SQL. Automated web scraping using BeautifulSoup and web APIs. Utilizing Pandas and NumPy for data wrangling, and Matplotlib, Folium, and Plotly to build interactive geospatial and statistical visualizations. Implementing regression, classification, and clustering algorithms using Scikit-learn. Navigating GitHub for version control. Leveraging GenAI tools to accelerate code generation, debugging, and data exploration.

Professional Engineer in Ontario 2018 - Current
PEO

- **Actionable Learnings:** As a licensed professional engineer, my foremost responsibility is to apply rigorous engineering methodology to ensure public safety in buildings and infrastructure projects.